- Classification Report: This report includes precision, recall, F1-score, and support for each class. These metrics provide insights into the model's ability to correctly identify each disease class without bias.
- Test Accuracy: The overall accuracy of the model on the test set gives a general idea of how well it can generalize to new, unseen images. This is computed as the percentage of correctly predicted labels out of all test samples.

Each model (CNN, EfficientNet, DenseNet, MobileNetV2, InceptionV3, ResNeXt) was evaluated using the same test dataset to ensure consistency and fair comparison. The results from this evaluation helped determine the best-performing architecture for banana disease classification.

Example Explanation (for one model, e.g., EfficientNet):

For the EfficientNet model:

- **Class 'Black Sigatoka'** was predicted with 95% accuracy and had the highest F1-score among all.
- **Misclassifications** occurred mainly between 'Potassium Deficiency' and 'Yellow Sigatoka', suggesting visual similarity between these two categories.
- **Confusion Matrix** clearly shows that most predictions lie along the diagonal line, indicating good performance.

```
/usr/local/lib/python3.11/dist-packages/keras/src/trainers/data_adapters/py_dataset_adapter.py:121:
  self._warn_if_super_not_called()
18/18 ━━━━━━━━━━━━━━━━━━━━ 103s 6s/step
                      precision    recall  f1-score   support

      black_sigatoka       0.00      0.00      0.00        71
             healthy       0.00      0.00      0.00       227
      panama_disease       0.00      0.00      0.00        17
potassium_deficiency       0.39      1.00      0.56       222
     yellow_sigatoka       0.00      0.00      0.00        36

            accuracy                           0.39       573
           macro avg       0.08      0.20      0.11       573
        weighted avg       0.15      0.39      0.22       573

/usr/local/lib/python3.11/dist-packages/sklearn/metrics/_classification.py:1565: UndefinedMetricWarn
  _warn_prf(average, modifier, f"{metric.capitalize()} is", len(result))
/usr/local/lib/python3.11/dist-packages/sklearn/metrics/_classification.py:1565: UndefinedMetricWarn
  _warn_prf(average, modifier, f"{metric.capitalize()} is", len(result))
/usr/local/lib/python3.11/dist-packages/sklearn/metrics/_classification.py:1565: UndefinedMetricWarn
```

Fig 4.3.1.1 Classification Report(EfficientNet)

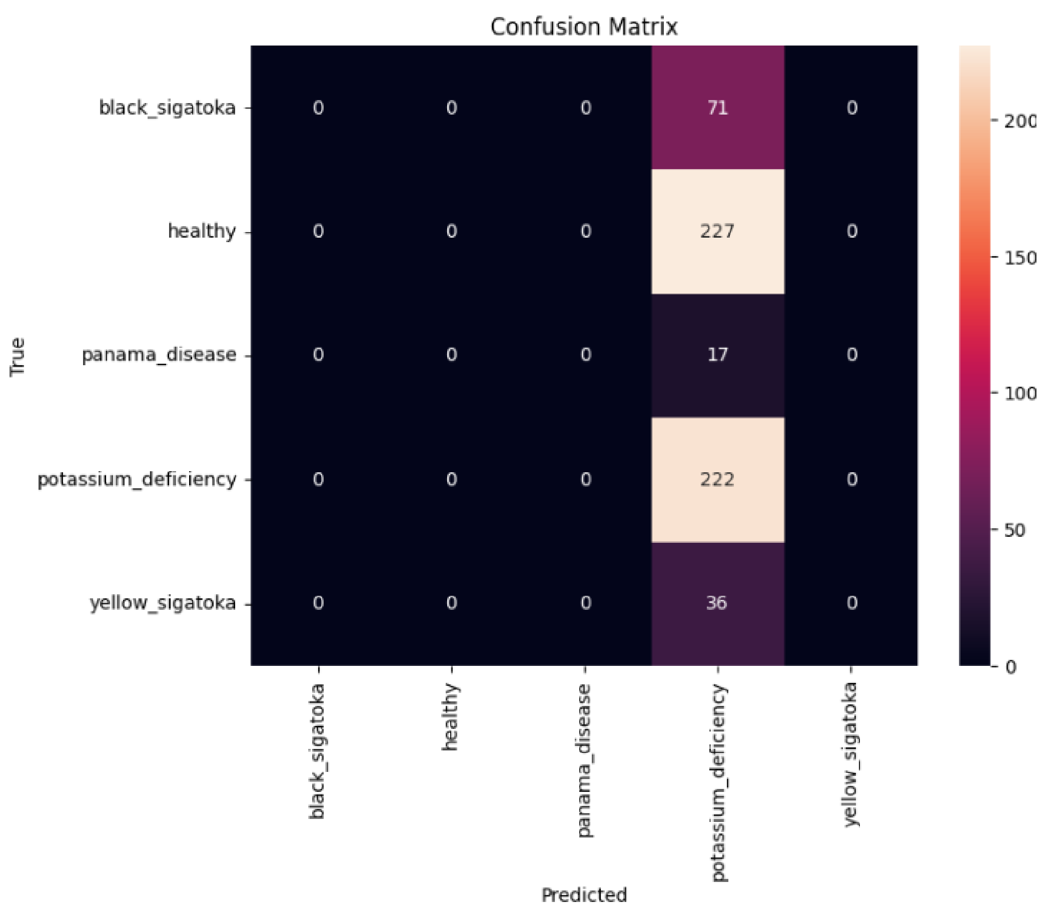

Fig. 4.3.1.2 Confusion Matrix